Exam 1: Part 1 (True/False and Multiple Choice). This part of exam 1 is not timed and is also open notes, open book, open Canvas materials, with no outside assistance permitted (from others or from outside materials).

Question 1 — 2 pts

The general purpose registers (GPRs) are not part of the data memory space (SRAM).

- ○ True
- ○ False

Question 2 — 2 pts

Assembler directives are commands to the assembler that are converted to machine code for program execution.

- ○ True
- ○ False

Question 3 — 2 pts

The general purpose registers (GPRs) are not part of the data memory space (SRAM).

- ○ True
- ○ False

Question 4 — 2 pts

The Instruction Register (IR) is a common register in microprocessor-based systems that contains the machine code for the current instruction.

- ○ True
- ○ False

Question 5 — 2 pts

A Harvard architecture microprocessor-based system uses a single address bus for both code and data memory. A Von Neumann architecture microprocessor-based system uses separate address and data buses for code and data memory.

- ○ True
- ○ False

Question 6 — 2 pts

Find the on-chip ROM size in k for the AVR chip with the address range 0x0000-0x1000.

- ○
- ○ 1
- ○ 32
- ○ 64
- ○ 16

Question 7 — 2 pts

Find the number of address lines for a 64k x 16 bits SRAM.

- ○ 16
- ○ 14
- ○ 6
- ○ 15
- ○ 5

Question 8 — 2 pts

The number of data pins (number of bits is each word) on a 64k x 16 bits SRAM..

- ○ 64
- ○ 24
- ○ 8
- ○ 16
- ○ 32

Question 9 — 2 pts

Find the capacity for a 64k x 16 bits SRAM..

Select the choice below for the capacity in terms of (number) k bits.

- ○ 64
- ○ 256
- ○ 1024
- ○ 512
- ○ 80

Question 10 — 1 pts

(True/False) The syntax for the following instruction is legal.

IN R32, PORTA

- ○ True
- ○ False

Question 11 — 1 pts

(True/False) The syntax for the following instruction is legal.

MOV R18, 0x10

- ○ True
- ○ False

Question 12 — 1 pts

(True/False) The syntax for the following instruction is legal.

LDS R12, 0X180

- ○ True
- ○ False

Question 13 — 1 pts

(True/False) The syntax for the following instruction is legal.

OUT R16, PORTB

- ○ True
- ○ False

Question 14 — 1 pts

(True/False) The syntax for the following instruction is legal.

LDI R16,255

- ○ True
- ○ False

Question 15 — 1 pts

The number of machine code **bytes** for the instruction LDS R31, 0x43.

- ○ 2
- ○ 1
- ○ 3
- ○ 4

Question 16 — 1 pts

The number of machine code **bytes** for the instruction ADD R2, R1.

- ○ 4
- ○ 1
- ○ 3
- ○ 2

Question 17 — 1 pts

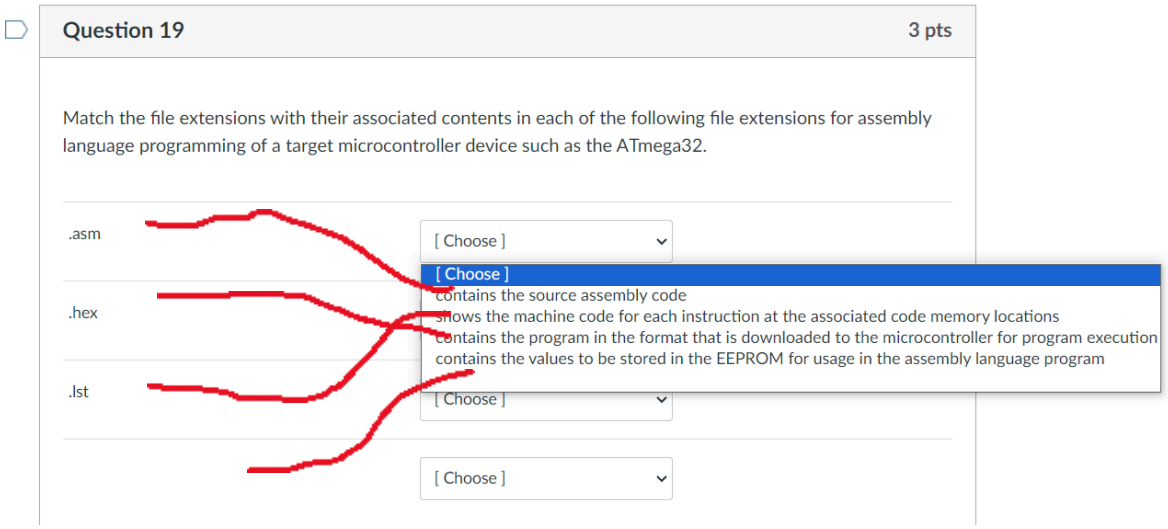
The number of machine code **bytes** for the instruction BRNE Again.

- ○ 2
- ○ 3
- ○ 1
- ○ 4

Question 18 — 1 pts

The number of machine code **bytes** for the instruction LDI R31, 0x43.

- ○ 4
- ○ 3
- ○ 1
- ○ 2



Match the file extensions on the left with what is contained in the file on the right. One 1 answer per file extension. Do not answer the last entry (blank).